DPhil Thesis Draft

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Contents

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List of Abbreviations

1-D, 2-D . . . One- or two-dimensional, referring in this thesis to spatial dimensions in an image.

Otter . . . . . One of the finest of water mammals.

Hedgehog . . . Quite a nice prickly friend.

Abbreviations

CNT carbon nanotube. 8

MW multi-walled. 8

SW single-walled. 8

1

Introduction

1.1 Motivation

Motivation

1.2 Graphene

1.2.1 crystal structure

Graphene has a hexagonal structure where the carbon atoms are hybridised sp^2, as detailed by the Pauling theory on chemical bonds. Based on the experimental evidence collected in the last 20 years, the model that seems to best describe the proprierties of graphene is the band-structure Wallace model, which dates back in 1947, and describes graphene as a semimetal with a linear dispersion of the electronic excitations. In the International Table of Crystallography, graphene is included in the symmorphic space group #191, with symmetry $D_6^1 h$ $(P6/mmm)$. The triangular lattice is obtained by considering two atoms per unit cell having the respective lattice vectors:

The carbon atoms in graphene form a two-dimensional hexagonal crystal with two atoms per unit cell, as shown in figure.

$$\mathbf{a}_1 = \frac{a}{2}(3, \sqrt{3}), \; \mathbf{a}_2 = \frac{a}{2}(3, -\sqrt{3}), \tag{1.1}$$

with $a \approx 1.42$ Å be the carbon-carbon bond length. Each atom belonging to one sublattice, called sublattice A, has three first neighbours belonging to the other sublattice, called sublattice B. With respect to one atom chosen from sublattice A as origin, the position of these first three neighbours in real space are given by:

$$\boldsymbol{\delta}_1 = \frac{a}{2}(1, \sqrt{3}),\ \boldsymbol{\delta}_2 = \frac{a}{2}(1, -\sqrt{3}),\ \boldsymbol{\delta}_3 = -a(1, 0), \tag{1.2}$$

The reciprocal-lattice vectors are then given

$$\mathbf{b}_1 = \frac{2\pi}{3a}(1, \sqrt{3}),\ \mathbf{b}_2 = \frac{2\pi}{3a}(1, -\sqrt{3}), \tag{1.3}$$

In order to define the vertices of the Brillouin zone one has to apply the Bragg condition:

$$2\mathbf{k} \cdot \mathbf{G} = 0 \tag{1.4}$$

where

$$\mathbf{G} = n\mathbf{b}_1 + m\mathbf{b}_2, \tag{1.5}$$

The first amongst these are:

$$\begin{aligned} &n = \pm 1, && m = 0, && \mathbf{G} = \pm \mathbf{b}_1, \\ &n = 0, && m = \pm 1, && \mathbf{G} = \pm \mathbf{b}_2, \\ &n = \pm 1, && m = \pm 1, && \mathbf{G} = \pm(\mathbf{b}_1 + \mathbf{b}_2), \end{aligned}$$

and all these vectors have the same modulus $|\mathbf{G}| = \frac{4\pi}{3a}$. By applying Bragg's condition in the first of the three cases we get:

$$\sqrt{3}k_x + k_y = \mp \frac{4\pi}{3a}, \tag{1.6}$$

which gets satisfied by the points:

$$K \equiv \mp \frac{2\pi}{3a}\left(\frac{1}{\sqrt{3}}, 1\right) \tag{1.7}$$

For the second case one gets:

$$\mp (\sqrt{3}k_x - k_y) = \frac{4\pi}{3a}, \tag{1.8}$$

which gives the points:

$$K' \equiv \pm \frac{2\pi}{3a}\left(\frac{1}{\sqrt{3}}, -1\right) \tag{1.9}$$

The third and last case gives the point:

$$M \equiv \pm\left(0, \pm \frac{2\pi}{3a}\right) \tag{1.10}$$

1.2.2 band dispersion

The electronic structure of graphene can be model with a tigh-binding hamiltonian. Following the second quantization formalism ($\hbar = 1$ units), the hamiltonian can be written as:

$$H = -t \sum_{\langle ij \rangle,\ \sigma} \left(a^{\dagger}_{\sigma,i} b_{\sigma,j} + \text{H.c.}\right) - t' \sum_{\langle\langle ij \rangle\rangle, \sigma} \left(a^{\dagger}_{\sigma,i} a_{\sigma,j} + b^{\dagger}_{\sigma,i} b_{\sigma,j} + \text{H.c.}\right), \tag{1.11}$$

The notation $\langle ij \rangle$ means that the sum runs over pairs of electrons which are nearest neighbour; similarly, the notation $\langle\langle ij \rangle\rangle$ specify a summation over the next nearest neighbours; the operators $a^{\dagger}_{i,\sigma}$ ($a_{i,\sigma}$) and $b^{\dagger}_{i,\sigma}$ ($b_{i,\sigma}$) creates (annihilates) an electron with spin σ ($\sigma = \uparrow, \downarrow$) for atoms on sublattice A and B, respectively. The value of t which reproduces best experimental result is $t \approx 2.8$ eV and quantify the nearest-neighbour hopping energy; similarly, t' denotes the next nearest-neighbour hopping energy. The dispersion energy computed from this model reads:

$$E_{\pm}(\mathbf{k}) = \epsilon_{\mathrm{p}} - t' f(\mathbf{k}) \pm t\sqrt{3 + f(\mathbf{k})}, \tag{1.12}$$

with $f(\mathbf{k})$ defined as:

$$f(\mathbf{k}) = 2\cos\left(\sqrt{3}k_y a\right) + 4\cos\left(\frac{\sqrt{3}}{2}k_y a\right)\cos\left(\frac{3}{2}k_x a\right). \tag{1.13}$$

Therefore, it is possible to identify an upper ($\pi*$) band associated with the plus sign and a lower (π) band described by the minus sign; these are symmetric with respect to the single-particle zero energy. The experimental estimate reported for next nearest-neighbour hopping energy is $t' \approx 0.1$eV thus the asymmetry introduced is small. It is worthwhile to notice that the function $3 + f(\mathbf{k})$

If the system is not doped, there is one electron per carbon atom, so two electrons per cell: it follows that the lower band is full and the upper band is empty the upper one is empty. The system is therefore a semi-metal. The peculiarities of the dispersion in graphene become apparent when analysing their trend near the point **K** and **K'**. By taking the positive components for the **K** as define in Eq. 1.7 and expanding the Eq. 1.12 as $\mathbf{k} = \mathbf{K} + \mathbf{q}$, with $|(\mathbf{q})| \ll |(\mathbf{K})|$, one gets:

$$E_{\pm}(\mathbf{q}) = \epsilon_p \pm \frac{3at}{2}|\mathbf{q}| \tag{1.14}$$

which is linear in $\mathbf{q}$. For intrinsic graphene, at $T = 0$K $E_F = \epsilon_p$. The Fermi velocity is defined by the usuale relation $v_F = \frac{1}{\hbar}\left(\frac{\mathrm{d}E}{\mathrm{d}\mathbf{k}}\right)$, and in graphene $v_F \simeq 1 \times 10^6 \mathrm{m\,s^{-1}}$.

A difference arises in the density of the states, when we limit ourselves to consider those near the Fermi energy. In the case of a two-dimensional electron gas, it is constant. On the contrary for graphene, the density of states per unite cell is:

$$\rho(E) = \frac{2A_{\mathrm{cell}}}{\pi}\frac{|E|}{v_F^2} \tag{1.15}$$

If the graphene is charge-neautral, the DOS has two Van Hove singularities at $\omega_{\mathrm{VH}} = \pm$ 1eV [1]. This behaviour is in stark constrat with the DOS of carbon nanotubes and clean graphene nanoribbons, as they show $1/\sqrt{E}$ singularities due to the 1D nature, which imposes a quantization perpendicular to the ribbon or tube longitudinal direction.

1.2.3 thermal conductivity

Thermal Properties of graphene

	Direct Gap			Indirect Gap		
	InP	Si	Ge	GaAs	InAs	Graphene
Lattice constant (Å)	5.87	5.43	5.65	5.65	6.06	2.46
E_g (eV)	1.35	1.12	0.66	1.42	0.36	0
μ (cm^{2}V$^-$1s$^-$1) at 300 K	2800	1400	3900	4600	16,000	200,000
v_{sat} (10^7 cm s^{-1})	2.2	1	0.6	2.2	4.0	>5
DOS (m^*/m_0)	0.077	1.08	0.56	0.067	0.023	0
ϵ_r	12.4	11.9	16.0	13.1	14.6	2.4
κ (W m^{-1} K^{-1})	68	150	60.2	46	27	5000

Table 1.1: Here is a caption

1.2.4 graphene as electrode material

can be used as electrode material

1.2.5 Graphene Transistors

The first graphene field-effect transistor (GFET) was the one reported by the Manchester group in 2004 where a scotch-taped exfoliated flake of graphene was deposited on top of 300 nm of SiO_2 acting as a back-gate [2]. The resulting output characteristics displayed were characteristics of an ambipolar device, with linear relation between the current and the drain-source voltage, without any saturation regime reached for the charge carrier [3] (Fig. **??**): this behaviour agrees with the semi-metal nature of graphene and indicates that transistor functionality was exclusively limited by inevitable contact resistances [4]. As for any transistor, the charge carriers can be varied by sweeping the gate voltage, which distinguishable branches of the gate-transfer characteristic; the charge neutrality point (Dirac point) is where they touch each other. Large positive gate voltages promote an electron accumulation in the channel (n-type channel), and large negative gate voltages lead to a p-type channel [5, 6].

Hysteresis between the forward and reverse sweeps have been reported depending on the sweeping rate and range [7], and attributed mainly to charge trapping in silanol groups (Si–OH), mediated and assisted by surface-bound H_2O molecules [8]. Despite the usefulness for its proof-of-concept purposes, back-gate devices suffered from parasitic capacitances, making impossible further integration with other components. Alternatives geometries have since then been explored, in order to increase the carriers mobility, to decrease contact resistances and parasitic capacitances, and to avoid Schottky barriers formation between the graphene and the semiconductor. Top-gate devices have been the first to chronologically appear in the literature. Back then in 2007, Lemme and co workeres used a stack of SiO_2 (20 nm), Ti (10 nm), and Au (100 nm) as top gate for modulating the current inside a large-area graphene channel transistor obtained by the exfoliation method [9]. Whilst the gate characteristics exhibit a similar trend of the characteristics (the top gate reduces the back-gate current modulation, lowering the drain current by one order of magnitude) and the hysteresis, weak current saturation at high drain-source bias have been reported [10]. This effect seem due to a partial charge pinning at the contacts at high back-gate voltages [**di2011charge**] and carrier drift velocity eventually saturating due to optical-phonon scattering [11]. Top-gated graphene MOSFET with CVD [**kedzierski2009graphene**] and epitaxial graphene have been also made [**kedzierski2008epitaxial**, **lin2010100**]; SiO_2, Al_2O_3, and HfO_2 have been used as top-gate dielectric.

However, despite the improvements obtained with a top gate, most graphene MOSFETs with large-area (gapless) graphene channels show on–off switching ratios of less than 10 at room temperature [12], much too little for complex logic circuits.

In 2012, Mehr et al. proposed a vertical arrangement of emitter (E), base (B), and collector (C) (like in a vacuum triode) where the graphene plays the base role and electrons travel vertically rather than transversely as in a GFET (Fig. **??**. When the device is in the ON state, the electrons tunnel with a Fowler-Nordheim tunnelling mechanism ($I \propto (V_{\mathrm{sd}}/d)^2 \ \exp\{-\alpha\ d/V_{\mathrm{sd}}\}$) through the emitter–base insulator (of thickness d), the base electrode (graphene), and the base–collector insulator. Vertical FETs based on hBN–graphene [**britnell2012field**] and WeS_2–graphene [13] heterostructures with ON/OFF ratios about 10^4 have been reported.

Solution methods have also been addressed, by which graphene is dropcasted on top of the wafer, and then spin-coated [14]. The main difficulties arise when attempting to quantify the structure and the quality of the deposited graphene, and not many methods can guarantee a good dispersion in organic solvents [**sun2011graphene**]. Simpler ways to obtain graphene oxide in solution have been reported [**van2016effect**], but once deposited, graphene oxide needs to be reduced for obtaining pure graphene – low mobility values (up to, at best, 1000 $cm^2\,V^{-1}\,s^{-1}$ [**cresti2008charge**]) caused by defects during the fabrication steps have discouraged further investigations along this direction.

Yet in spite of the very different physics and resulting DC electrical behavior, it has been shown that graphene FETs behave very much like their semiconductor counterparts when they are operated at high frequency, under alternating-current (AC) conditions. This comparison is made by measuring each system's frequency response, its output signal strength when it is given a particular input signal: features such as ballistic transport, linear current-voltage characteristics, high transconductance, and very high ($> 10^8\ A\,cm^{-2}$) sustainable currents are all ideal features for analog/RF applications, which do note need to be switched off [**meric2008rf**, 11].

The primary means of estimating frequency response is through the cutoff frequency: it is a fundamental property indicating how quickly a signal can travel from the gate to the drain of the device. The frequency response depends inversely on the length of the gate, and proportionally to the transconductance, which is loosely proportional to mobility. Thus, to make faster devices, we can either reduce the gate length or use higher mobility materials [12].

The first graphene MOSFET with a cutoff frequency f_T in the gigahertz range has been reported in late 2008 [**meric2008rf**]. In 2009, graphene MOSFETs with 350-nm gates and a cutoff frequency of 50 GHz have been presented [**lin2009development**]; in early 2010, a graphene MOSFET with a 240-nm gate showing an f_T of 100 GHz has been reported [**lin2010100**], soon followed by a 144-nm gate graphene MOSFET with f_T ¼ 300 GHz later that year [**liao2010high**]. Finally, in 2012, a graphene MOSFET with a 40-nm-long gate showing 350-GHz f_T [**wu2012state**] and a 67-nm gate graphene MOSFET with an f_T of 427 GHz have been reported [**cheng2012high**]. Graphene RF MOSFETs have successfully been realized using exfoliated (e.g., [**liao2010high**] and [**lin2009development**]), epitaxial (e.g., [**lin2010100**] and [10]), and CVD ([**wu2011high**]) graphene.

Fig. **??** shows the cutoff frequency of the best graphene MOSFETs reported, together with the f_T performance of competing RF FET types. The record f_T data of the competing FETs are 688 GHz for a 40-nm GaAs mHEMT[**kim2011f**] and 644 GHz for a 30-nm InP HEMT [**kim201030**], 485 GHz for a 29-nm Si MOSFET [**lee2007record**], 152 GHz for a 100-nm GaAs pHEMT [**nguyen1989characterization**], and 153 GHz for a 100-nm carbon nanotube (CNT) MOSFET [**steiner2012high**]. Worthwhile to mention also the results obtained by Han et al. [**han2014graphene**] in 2014, when they implemented a high-performance three-stage graphene integrated circuit using Si fabrication steps and CVD graphene: the reported performance for f_T and $f_{\max}$ were ranging from 8 to 10 GHz and from 8.5 to 13.5 GHz.

1.3 Carbon-based Field Effect Transistor

A great deal of attention from the scientific community has been dedicated – since the late 90s – to carbon nanotubes (CNTs) [15]. This crystalline form of carbon is a rolled up sheet of graphene and comes in two major flavours, namely single-walled (SW) and multi-walled (MW). With typical diameter ranging from 1 to 100 nm, these hollow cylindrical structures display an electronic band gap which depends on the chirality and the diameter of the nanotube [16].

The chiral vector is the direction along which the graphene sheet is rolled up and becomes the circumference of the nanotube: it must be a linear combination of the basis vectors with integers coefficients (m, n). Based on the values of m and n, CNTs can be metallic ($m = n$), quasi-metallic ($m - n$ is a multiple of 3, with $m \cdot n \neq 0$ and $m \neq n$), and semiconducting for the majority of the remaining cases – exceptions due to the curvature (spin–orbit couplings) in small-diameter tubes are also known [17].

Detail here the carbon nanotube journey.

Device type	I_{on}/I_{off}	SS (mV/dec)	g_{mt} (µS cm^{-1})	L_{ch} (nm)	f_T [f_{max}] (GHz)	Ref.
Bottom gate	10^4	94	~ 40	9	—	[18]
	10^5	~ 85	40	15	—	[19]
	10^5	~ 85	20	20	—	[19]
	10^5	60	20	200	—	[20]
	10^5	~ 63	—	200	—	[21]
	10^4	80	50	240	—	[22]
	—	—	143	800	[1]	[23]
	10^4	—	—	1,500	1.5 – 3.0	[24]
	—	70	$12^{\dagger}$	2,000[1]	—	[25]
	10^8	—	—	$< 10,000$[1]	—	[26]
	—	—	- 500	—	30	[27]
Top gate	10^6	—	20[2]	80	—	[28]
	2	—	~ 50	100	25	[29]
	50	—	310	100	~ 70 [80]	[30]
	—	—	0.3	210	—	[31]
	10^5	—	33.5	85,000	—	[32]

Table 1.2: Figures of merit for a few CNFET reported in literature. S: subthreshold swing; g_m: intrinsic transconductances; L_{ch}: channel length; μ: carriers mobility

†Normalized by the tube width.
[1] Referred to the gated length of a nanotube.
[2] For the n-type transistor, whereas 10 µS were measured for the p-type.

Device type	I_{on} (A)	I_{off} (A)	I_{on}/I_{off}	g_{mt} (µS)	SS (mV/dec)
Conventional	$1.41{*}10^{-5}$	$4.06{*}10^{-10}$	$3.4{*}10^4$	29	99.6
Partial S/D-doped	$1.50{*}10^{-5}$	$2.33{*}10^{-8}$	$6.4{*}10^2$	63.6	93.1
Schottky-Barrier	$3.26{*}10^{-6}$	$2.31{*}10^{-8}$	$1.41{*}10^2$	19.6	109.8
Tunnel	$5.53{*}10^{-6}$	$1.16{*}10^{-8}$	$4.76{*}10^2$	13	107.9

Table 1.3: Comparative simulation for the four major type of CNFETs as reported in [33]. These results are obtained by setting $V_{sd} = 0.5$ V.

1.4 Graphene Nanoribbons

The scientific interest for graphene nanoribbons started in the physics community. Although this may appear irrelevant at the first glance, we will see that this actually had an impact over the last twenty years of progress in the field. The physics community was looking for a way to open a bandgap in graphene. As mentioned in Sec. 1.1, graphene is a semi-metal, therefore unsuitable for all those semiconducting applications where the conduction paths for the charge carriers are well defined.

The first class of methods for synthesising graphene nanoribbons are methods that involve cutting or producing rectangular shapes from the graphene lattice through macroscopic methods. For this reason, they are also known as *top-down* approaches. Although they have seen an improvements in recent years, common drawbacks for this class of methods are a random distribution in the widths, a non-atomical control of the edge structures, and non-negligible impurity concentrations left after the process.

Kim and co-workers synthesized GNRs by employing the electron-beam lithography and etching technique [34]. The widths of GNRs obtained by this technique were scattered over a broad range, and most of the edges were extremely rough. An 8 nm GNR FET showed a dramatically improved on/off ratio of about 160.

Nanocutting of graphene is another promising method of GNR synthesis. In 2008, Strachan et al. used thermally activated Fe nanoparticles as chemical scissors to cut graphene sheet along a specific crystallographic direction [34, 35]. Later, Ni and other catalysts were employed as well The carbon-carbon bond breaking is driven by the combustion reaction $\mathrm{Ni} + \mathrm{C_{graphene}} + 2\,\mathrm{H_2}\uparrow \longrightarrow \mathrm{Ni/Fe} + \mathrm{CH_4}\uparrow$ catalysed by the nanoparticles, thus producing hydrocarbon compounds. One of the reason that have discouraged these synthetic way is the random width distribution obtained: this is likely to be a consequence of the change in direction when nanoparticles encounter defects or edges.

It will be interesting to discuss the $E_\mathrm{g} \propto 1/W$ relation as the model used for describing gnr is the particle in the box, but in the toy model (particle in a box) as a relation of the kind $E_\mathrm{g} \propto 1/W^2$.

1.5 Delocalised π-electron Systems

Porphyrin

2
Foundations

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2.1 Charge Transport In Zero-Dimensional Objects

quantum dots transport physics

2.2 Charge Transport in One-Dimensional Objects

nanowires transport physics

2.3 Thermoelectric Phenomena in Zero-Dimensional Structures

What are the necessary changes that one has to take into account when dealing also with other extensive variable such as a temperature gradient

2.4 Density Functional Theory

What are the relevant parameters/quantities that we are interested to calculate ab initio Why we opted for a pseudopotential such as SIESTA rather than others (quantum espresso). Explain why calculating the isolated gas phase all-electron electronic structure is legit for a qualitative see how the electron density is distributed over the system.

Remember also to emphasise the difference from stricly and non-stricly localised functions used for computing the electronic properties: when the functions are stricly localised they are zero outside a given radius from the atom they are associated with and so the radius is a key quality determined parameter for the basis set

2.5 Non-Equilibrium Green Function Theory

This section is needed to explain formally the results from transiesta and TB-trans calculations

3
Instrumentations

Contents

3.1 Room Temperature DC Measurements

Describe here the probe station setup]

3.2 Dipstick DC Measurements

Describe here the dipstick used for both liquid N_2 and liquid He

3.3 Cryogenic DC Measurements

Describe here the delft box. Describe how the delft box modular structure enables to adapt two and multi-terminal DC measurements

3.4 Cryogenic AC Measurements

Describe here the delft box + lock-in setup and why it is so complicated to set it usepackage

3.5 Thermoelectric Measurements

Describe here the setup used for the thermoelectric devices and why it needs a dedicated section Describe also here from a setup point of view which complications have raised during the setup (direct lines instead of dc lines, the need to use nano-D connectors and to design the a new pcb board).

4
Fabrication

Contents

4.1 General Fabrication Procedure

The steps are well known: chip preparation, e-beam, development, metal deposition and lift-off, graphene transfer old process, graphenea graphene

4.1.1 Graphene Preparation and Wet Transferring

Initially the graphene was transferred on top of the substrate through the wet transfer method. The graphene was supplied by Graphene ain 5 x 5 cm2 rectangles, with a PMMA front-layer and a back-layer of partially removed graphene. Before transfer, the back-layer had to be fully removed via etching, and the remaining stack appropriately sized for the substrates for this work. The graphene arrived in a vacuum-sealed pack, which had to be transported to the cleanroom before being opened and the following preparation steps initiated. Figure 6.2 depicts the main stages involved in preparing the graphene for transfe

4.2 Graphene Nanogaps

Here three methods have been exploited: standard feedback electroburning, gate-dependent electroburning, and litographic patterning of graphene nanogap

4.3 Chip Design

Here the designs for the different chips that have been used. Give a proper credit to Dr. Alex Gee for the fabrication of thermoelectric and the common gate ones.

4.3.1 2T Design

How convenient and 'quick' is the usage of a backgate device. List, as usual, pro and cons

4.3.2 3T Design

Why the need to use 3T over 2T

Local Gates from Waterloo

Give proper credit

Interconnected Drain and Gate

The one I made and also made with the help of Alex

4.3.3 Thermoelectric Device

Common and different features of the thermoelectric

4.4 Deposition Process and Related Phenomna

Why it has been important to try different deposition techiniques How this has been qualitatevely addressed.

5

Room Temperature Graphene Nanoribbon Single-Electron Transistor

Contents

5.1 Introduction

A brief intro why single-electron transistors with gnr are interesting A brief intro to what we are gonna do/measure in this chapter

5.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters) Explained why the scarsity of characterisation checks is intrinsically scarse in molecular electronics.

5.3 Electrical Transport Measurements

What type of experiments have been performed and which tricks have been used in triton to do so. What are the evolution in temperature of the zero-bias, low-bias and high bias conductance. Look at the dependence in temperature and explain possible interpretations: why it is not a Luttinger Liquid (the gnr has a bandgap) and what possible phonon modes are responsible at some point for desctructing the

5.4 Summary

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Chemical Doping of the Bandgap for Cove Graphene Nanoribbons

Contents

6.1 Introduction

A brief intro on how why it is important to study this

6.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters) Explained why the scarsity of characterisation checks i s intrinsically scarse in molecular electronics.

6.3 *Ab-initio* Experiments

What DFT and NEGF have been used and for what.

6.4 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest badgap distance.

6.5 Summary

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Thermocurrent Spectroscopy of cove Graphene Nanoribbons

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7.1 Introduction

A brief intro on how why it is important to study this what has been done.

7.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters) Explained why the scarsity of characterisation checks i s intrinsically scarse in molecular electronics.

7.3 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest badgap distance.

7.4 Summary

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Transport Spectroscopy of Biaryl-linked Dy–Tb Porphyrin

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8.1 Introduction

A brief intro on how why it is important to study this

8.2 Sample Preparation and Characterisation

How the sample it has been prepared (lot of crossreference with previous chapters) Explained why the scarsity of characterisation checks i s intrinsically scarse in molecular electronics.

8.3 Electrical Transport Measurements

Explain how the calculations have initially suggested the comparison between the base case and the methoxy groups, as they have the largest badgap distance.

8.4 Summary

9 Conclusions and Outlook

Contents

9.1 Conclusions

Here the final conclusions

9.2 Outlook

Appendices

Cor animalium, fundamentum eft vitæ, princeps omnium, Microcofmi Sol, a quo omnis vegetatio dependet, vigor omnis & robur emanat.

The heart of animals is the foundation of their life, the sovereign of everything within them, the sun of their microcosm, that upon which all growth depends, from which all power proceeds.

— William Harvey [36]

A
Review of Cardiac Physiology and Electrophysiology

Appendices are just like chapters. Their sections and subsections get numbered and included in the table of contents; figures and equations and tables added up, etc. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed et dui sem. Aliquam dictum et ante ut semper. Donec sollicitudin sed quam at aliquet. Sed maximus diam elementum justo auctor, eget volutpat elit eleifend. Curabitur hendrerit ligula in erat feugiat, at rutrum risus suscipit. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Integer risus nulla, facilisis eget lacinia a, pretium mattis metus. Vestibulum aliquam varius ligula nec consectetur. Maecenas ac ipsum odio. Cras ac elit consequat, eleifend ipsum sodales, euismod nunc. Nam vitae tempor enim, sit amet eleifend nisi. Etiam at erat vel neque consequat.

A.1 Anatomy

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A.2 Mechanical Cycle

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A.3 Electrical Cycle

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A.4 Cellular Electromechanical Coupling

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The first kind of intellectual and artistic personality belongs to the hedgehogs, the second to the foxes . . .

— Sir Isaiah Berlin [37]

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